KØBENHAVNS UNIVERSITET

Bachelorprojekt i Matematik-økonomi

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American Options

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Abstract

Her skriver I et kort resumé på både dansk og engelsk af jeres projekt. Hvert resumé skal ikke fylde mere end en halv side.

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1 Introduction

Financial derivatives, and option pricing in particular, play an important role in modern financial markets through risk management and portfolio optimization. Accurate pricing and hedging of options is therefore essential and a relevant problem. European options can often be valued analytically under the Black–Scholes framework, however American options present additional challenges due to the possibility of early exercise, making numerical methods necessary to investigate.

2 Theory

2.1 Options

2.1.1 European options:

A financial contract is a financial derivative, or contingent claim, if the value at maturity T is determined by the underlying asset at time T. The simplest types being the call and put options. A European call option is a financial contract where the holder has the right but not the obligation to buy the underlying asset at a stated price at a specific time. A European put option is a financial contract where the holder has the right but not the obligation to sell the underlying asset at a stated price at a specific time. This price is called the strike price and the specific time is the maturity of the derivative. Past maturity the option expires becoming worthless. These types of contingent claims with stock price S and strike price K have the payoff function given:

$$\textbf{EU Call} = \max\{S - K, 0\}, \quad \textbf{EU Put} = \max\{K - S, 0\}.$$

The payoff function for the European call and European put ensure that the option holder only exercises if the payoff function is positive, meaning the option is in the money(ITM). If the payoff function on the other hand is less than or equal to zero, the option holder will let the option expire and the option will be out of the money(OTM).

2.1.2 American options:

In contrast the American option has the characteristic that it can be exercised at any time from purchase to maturity, so for $t < T$. The American call option will have the same characteristics as the European call. For the American put, the holder has the right to sell the underlying asset at strike price K for any $t < T$. The payoff function for the exercise value of the American put at any time $t < T$ is:

$$\mathbf{V(S,t)} = \max\{(K - S_t),\ \text{continuation value}, 0\}$$

Det her er ikke en payoff function, da du ikke kan få continuation udbetalt til tid t
Value function er mere korrekt

If the option holder exercises the option, they loose the possibility of exercising later, which could give a higher expected exercise value. Therefore the goal is to find the optimal time to exercise the option, to maximize the expected discounted payoff.

2.2 Valuation algorithm (section 2 Longstaff)

We assume the the valuation scheme is based on the assumptions in Section 2 of Longstaff and Schwartz (2001). The underlying filtered probability space is given by $(\Omega, \mathcal{F}, P)$ over a finite time horizon of $[0, T]$. The time horizon $[0, T]$ can be discretized, such that the option has early exercise times of

$$t_0 = 0 < t_1 \leq t_2 \leq ... \leq t_N = T.$$

Where the discretization steps can be interpreted as the possible exercise opportunities for the given financial contract.

The sample space Ω contains all possible realizations of the stochastic economy over the time period $[0, T]$. Where $\omega_i \in \{\omega_1, ..., \omega_I\}$ are the different sample paths in Ω.

The sigma-algebra $\mathcal{F}$ represents the distinct events at time T, with P being the probability measure defined on the elements of $\mathcal{F}$. Furthermore, $F = \{\mathcal{F}_t : t \in [0, T]\}$ is defined to be the filtration generated by the relevant price process for the assets in the underlying economy, representing the information that is available over time. It is assumed that $\mathcal{F}_T = F$.

Equivalent to the no arbitrage theory, the existence of a risk-neutral measure for the economy $\mathcal{Q}$ is assumed. Under $\mathcal{Q}$, all assets are assumed to earn the risk-free rate in expectation, implying the discounted prices follow a martingale process. $\mathcal{Q}$ is a transformed probability measure, where contingent claims can be consistently valued as the discounted expectation of their future payoffs and preserving the same null events as the physical probability measure P. In addition to this it is assumed that the risk-neutral stock price process of the underlying asset follows the stochastic differential equation (SDE):

$$dS_t = rS_t dt + \sigma S_t dW_t$$

Here W_t is a standard Brownian motion, which is defined with the following properties:

$$W_0 = 0 \; a.s., \quad W_t - W_s \sim \mathcal{N}(0, t-s), \forall s < t, \quad W_{t_0} \perp\!\!\!\perp W_{t_1} - W_{t_0} \perp\!\!\!\perp W_{t_n} - W_{t_{n-1}}, \forall t_0, ..., t_n \in [0, T]$$

2.3 Geometric Brownian Motion (GBM)

A Brownian motion is a continuous stochastic process used for modeling the uncertainty and random fluctuations observed in financial markets. We wish to obtain an analytical solution to the SDE to simulate the behavior of the Brownian motion. It is therefore assumed that there exists a solution to this SDE. By defining $Z_t = lnS_t$, we can use Itô's lemma to derive the expression for S_t. The solution is then given by Björk (2020) chapter 5:

Forvirrende at have dem alle side om side.

Ifht uafhængighed er det skrevet uklart op. Du påstår uafhængighed mellem de tre par, ikke for generelle inkrementer, enten sæt ... mellem de (t_1-t_0) og (t_n-t_{n-1}), eller skriv at inkrementer er uafhængige i ord.

I praksis simulerer du Brownian increments, ikke værdien af Brownian motion til tid t.

En mere præcis beskrivelse ville være:

W(t+ \Delta) t) - W(t) = \sqrt{\Delta t} \cdot Z

Man priser under Q målet, hvor drift er r. Skriv et sted i 2.2 at man priser under risk neutral measure Q, og så er det bare den du priser under i resten af opgaven

$$S_t = S_0 \cdot exp\left\{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W(t)\right\}$$ skal det være mu eller r

\exp i latex

Here μ is the drift of the Brownian motion and σ is the diffusion. We want to model the movements of a Brownian motion, which is a continuous stochastic process. To do so the process has to be discretized to a finite number of steps, $0 = t_0 < t_1 < \cdots < t_n = T$. When the discretizations are done, $W(t)$ can also be written as:

$$W(t) = \sqrt{\Delta t} \cdot Z$$

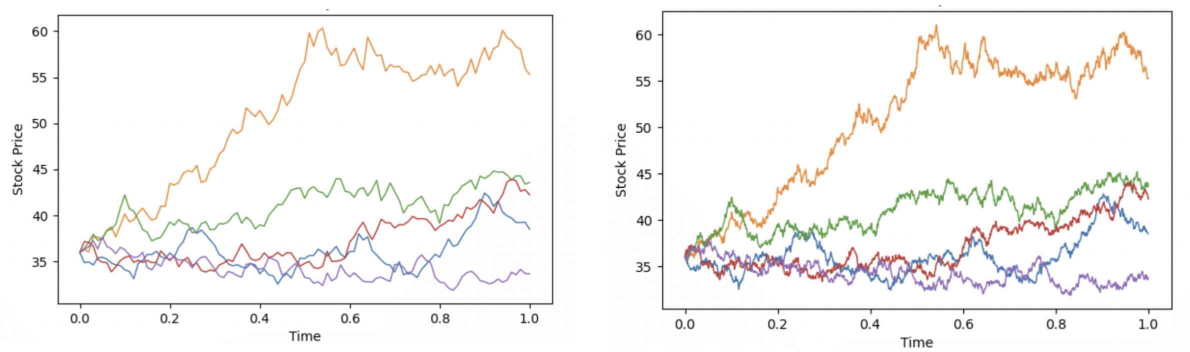
where $Z \sim \mathcal{N}(0, 1)$. The stock prices can then be simulated the stock price of the underlying asset by:

r

$$S_t = S_0 \cdot exp\left\{\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z\right\} \tag{2.1}$$

With this equation, and the N discretized time steps being the early exercise times with length $\Delta t = dt = \frac{1}{N}$, the price paths for a geometric Brownian motion can be plotted.

I virkeligheden er det S_{t+\Delta t} = S_t * \exp(...) Du viser kun første skridt her



(a) 100 steps

(b) 1000 steps

Figure 1: Brownian motion simulated stock paths

Price paths simulated with $S_0 = 36, r = 0.06$ and $\sigma = 0.2$

Figure 1 shows how the same parameters, simulated stock price paths can vary due to the stochastic feature of a Brownian motion. When increasing the number of steps the plot shows a finer discretization of the process making short-term fluctuations more visible.

2.4 Black-Scholes model

To acquire a mathematical model for option pricing, we introduce the Black-Scholes framework. The derivation of the Black-Scholes partial differential equation (PDE) follows the arbitrage pricing approach in Björk (2020) - chapter 7.

In the Black-Scholes model, it is assumed that the market consists of two assets, the stock price being the risky asset. A derivative security $V(t, S_t)$ with underlying stock S_t with maturity T is considered. Here the underlying stock price is assumed to follow a geometric Brownian motion.

Since $V(t, S_t)$ is a function of a stochastic process, Itô's lemma can be applied to determine the dynamics of the derivative price. With Itô's lemma we get:

$$dV = \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S\frac{\partial V}{\partial S} dW_t$$

To derive the arbitrage free price of V, we construct a self-financing portfolio Π consisting of one option V and a short position of Δ units of the stock. Here the value of the portfolio will be:

$$\Pi = V - \Delta S.$$

In each time interval the change in the value can be noted : $d\Pi = dV - \Delta dS$.

It is known that the restriction $w^S + w^F = 1$ must hold. $w^S\sigma + w^F\sigma_F = 0$ also applies in the risk-neutral valuation scheme. The values of dV and dS can then be substituted into the equation of $d\Pi$ to get:

$$\begin{aligned} d\Pi &= \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S\frac{\partial V}{\partial S} dW_t - \Delta(rSdt + \sigma S dW_t) \\ &= \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2} - \Delta rS\right) dt + \left(\sigma S\frac{\partial V}{\partial S} - \Delta\sigma S\right) dW_t. \end{aligned}$$

Because of the application of the linear equation $w^S\sigma + w^F\sigma_F = 0$, we can define $\Delta = \frac{\partial V}{\partial S}$. This will eliminate the stochastic term of $d\Pi$ giving the dynamics:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2}\right) dt.$$

Since the portfolio is in absence of arbitrage, we know that it must earn the risk-free rate r, giving $d\Pi = r\Pi dt$. Knowing $\Pi = V - \Delta S$ and $d\Pi$, these can be substituted in. Reducing the expression to get the PDE:

$$d\Pi = r\left(V - \frac{\partial V}{\partial S}S\right) dt$$

$$\left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2}\right) dt = r\left(V - \frac{\partial V}{\partial S}S\right) dt$$

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

For American options early exercise is available and the Black-Scholes PDE therefore does not hold with equality. It is possible for the option-holder to either hold the option, which will result in the PDE holding with equality. Otherwise the option value coincides with the payoff in the exercise region, meaning that the holder should exercise the option now. The Black-Scholes PDE for an American option will then be:

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV \leq 0.$$

The boundary conditions for an American options are then:

$$V(S,T) = h(S), \quad V(S,t) \geq h(S)$$

with h being the payoff function of the underlying asset at maturity.

When looking at American options the possibility of early exercise introduces a free boundary problem, since the region in which early exercise is optimal versus the region to hold the option is not known in advance and must be determined as part of the solution. The inequality reflects the early exercise feature, ensuring that the option value is at least as large as the payoff. Since the holder wishes to maximize the payoff, the optimal stopping boundary will be at $V(S,t) = h(S)$.

As a consequence, there is no explicit solution for American options, and pricing them is then a problem of solving for both the option value and finding the optimal stopping time, and the optimal exercise strategy is therefore needed.

2.5 Optimal stopping

To price the American options, we formulate it as a problem of finding the optimal exercise strategy. The method follows Chapter 28 in Björk (2020). A stopping time represents a random time at which a decision is made, based only on the information available up to that time. For $(\Omega, \mathcal{F}, P)$ defined as in Section 2.2, a non-negative random variable τ will be a stopping time with respect to the filtration F if:

$$\{\tau \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0.$$

The stopping time τ is restricted to fall on one of the time steps made from the discretization $0 = t_0 < t_1 < \cdots < t_n = T$. For any time point n and a given stopping time τ, with $n < \tau < T$ the value process is defined by: $E[Z_\tau|\mathcal{F}_n]$ with Z being an integrable process. From this we can define the optimal value process:

$$V_n = \sup_{n<\tau<T} E[Z_\tau|\mathcal{F}_n] = E[Z_{\hat{\tau}_{n,x}}|\mathcal{F}_n].$$

The optimal value process is the process that finds the supremum of the value process by choosing τ. Where $\hat{\tau}_{n,x}$ will denote the optimal stopping time. The stock price is defined so it follows a Geometric Brownian Motion (GBM), which a Markov process. A Markov process has the property that the future development only depends on the current state, therefore it is only necessary to compare he current state with the following time. There are then three different stopping strategies to compare:

Strategy 1: Use the optimal stopping strategy and stop at $\hat{\tau}_n$.

Strategy 2: Stop immediately.

Strategy 3: Waiting until time t_{n+1}, then start using the optimal strategy using $\hat{\tau}_{n+1}$.

Following the definition of $\hat{\tau}_n$ as well as the laws of conditional expectations, the values of the three strategies at time $t = t_n$ are:

$$\textbf{S1}: V_n, \quad \textbf{S2}: Z_n \text{ and } \quad \textbf{S3}: E[Z_{\hat{\tau}_{n+1}}|F_n] = E[E[Z_{\hat{\tau}_{n+1}}|F_{n+1}]|F_n] = E[V_{n+1}|F_n].$$

Since the first strategy is the best by definition, the following inequalities will hold:

$$V_n \geq Z_n \tag{2.2}$$

$$V_n \geq E[V_{n+1}|\mathcal{F}_n] \tag{2.3}$$

At time $t = t_n$ there are two choices, either stopping now resulting in S2 or continuing which is S3. Therefore either equation 2.2 or 2.3 must hold with equality. Since one of the inequalities must hold by equality, the optimal value process can be written as:

$$V_n = max\{Z_n, E[V_{n+1}|\mathcal{F}_n]\}. \tag{2.4}$$

Where V_n is the solution to the backward recursion defined by 2.4 and $V_T = Z_T$. This defines the optimal stopping strategy, since $V_n = Z_n$ if and only if it is optimal to stop at time n and $V_n > Z_n$ if it is optimal to continue. Hence the optimal stopping strategy will result in stopping the first time the value process equals the payoff process. The optimal stopping time at time $t = 0$ and $t = t_n$ can then be defined as:

$$\hat{\tau} = min\{n \geq 0|V_n = Z_n\}, \quad \hat{\tau}_n = min\{k \geq n|V_k = Z_k\}.$$

Together this creates the necessary framework to approach the problem of pricing an American option recursively. When approaching the pricing problem recursively, it is enough to look at the exercise value and the continuation value.

The continuation region of the discrete time problem is then defined as $C = \{t \geq 0|V_t > Z_t\}$. Letting $N \to \infty$ the problem approaches the continuous time problem. In continuous time the three conditions hold:

$$\hat{\tau} = min\{t \geq 0|V_t = b_t\},$$

$$V_t > max(K - s, 0), (t, s) \in C,$$

$$V_t = max(K - s, 0), (t, s) \in C^c.$$

Where b_t is the continuously differential boundary of C, meaning that $b \in C^1$ and $(t, b_t) \in \partial C$.

For an American call option paying no dividends and with a positive interest rate, the early exercise will never be preferable. Therefore the American and European call prices are identical, making it possible to use the Black-Scholes formula for pricing an American call option.

For an American put the early exercise can be preferable sometimes. The decision to exercise relies only on time t and the underlying stock price S_t. There will exist an early exercise boundary, where the first time the underlying reaches the boundary, it is optimal to exercise. This boundary can be viewed as a function of t which is increasing as you approach maturity. The function is increasing because the chance of the underlying stock price going below the exercise boundary decreases as it approaches maturity. The theoretical exercise boundary together with simulated stock price paths is visualized in Figure 2.

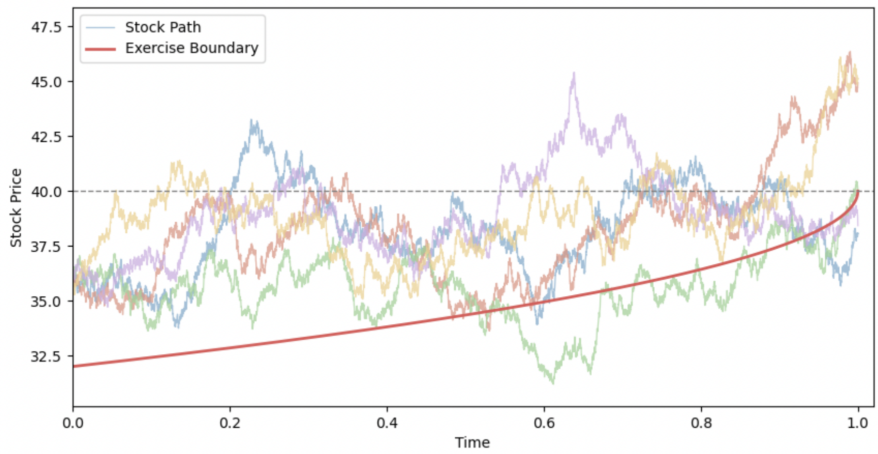


Figure 2: Optimal stopping boundary

Simulated price paths and the theoretical exercise boundary simulated with $S_0 = 36, r = 0.06$ and $\sigma = 0.2$.

In Figure 2, the optimal exercise boundary indicates the conditions where early exercise of the American put option is optimal. Simulated stock paths crossing below the boundary enter the exercise region, while paths remaining above the boundary suggest it is optimal to continue holding the option. Illustrating that early exercise is not always optimal.

Er det skrevet eksplicit at der ikke er en analytisk måde at prisfastsætte de amerikanske put optioner?

3 Options pricing models

Since no analytical solution exists for pricing an American put option, two alternative numerical methods are introduced. Both methods are based on the established optimal stopping framework. A key feature in the numerical pricing of American options is the discretization introduced in Section 2.2. Specifically, when implementing the two models, the American option is approximated by a Bermudan option, where early exercise is restricted to a finite set of predetermined exercise dates. The optimal stopping strategy is then applied at these discretization points.

3.1 CRR Binomial model

First we introduce the Cox, Ross and Rubenstein (CRR) method, to implement a reliable theoretical benchmark to compare the Least Square Monte Carlos Model to. The setup for this model is based on Section 3 in the Cox, Ross, and Rubinstein (1979) paper. In the model, it is assumed that the stock price follows a multiplicative binomial process over a finite number of periods. So the stock price will either go up with probability q or go down with probability $1-q$, resulting in the stock price at $t=1$ being $S_0 \cdot u$ or $S_0 \cdot d$. For the binomial model to hold, it is given that $d < r < u$, to ensure there is no opportunities for profitable riskless arbitrage. The values of u, d and q are given by:

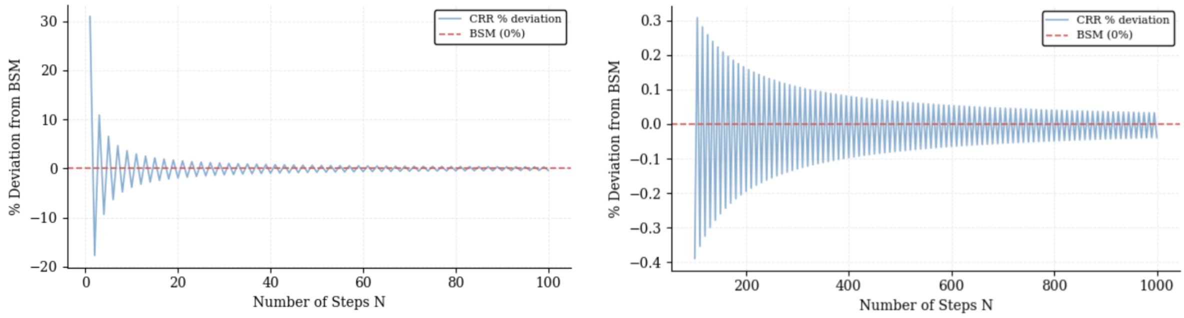
$$u = e^{\sigma\sqrt{dt}}, \quad d = e^{-\sigma\sqrt{dt}}, \quad q = \frac{r-d}{u-d}.$$

When calculating the price for the underlying stock, the recursive method is used. The stock price will be the expectation of the discounted future values under the risk-neutral measure Q. When finding the price of the underlying option, either put or call, we start by finding the value of the underlying stock for each node at time T. The prices found will be the expected exercise value at the given time.

For $t = T-1$ we discount the expected value from time T. The early exercise value is compared with the continuation value to see if early exercise is optimal in any of the calculated nodes for time $t = T-1$. If early exercise is optimal, the option is exercised. If early exercise is not optimal, we move to time $t = T-2$, here we again discount the calculated values from the previous time step and compare to see if the early exercise now is preferable. This continues until time step $t = 0$. This can be expanded for an infinite amount of time steps N.

An important property of the CRR model is its convergence to the Black–Scholes price for European options as the time discretization becomes finer. More specifically, as the number of time steps N increases and the time increment $dt = \frac{1}{N}$ approach zero, the discrete-time stock price dynamics in the CRR model converges to the continuous-time dynamics assumed in the Black–Scholes framework.

Since the CRR model both accommodates the early exercise feature which the Black–Scholes framework is not able to incorporate, the CRR model is both as a consistent approximation to Black–Scholes in the European case and a practical numerical method for pricing American options. The CRR convergence can be illustrated by plotting the deviation between the CRR and Black–Scholes European put price.



(a) N∈[0,100]

(b) N∈[100,1000]

Figure 3: CRR convergence plots

The convergence of the CRR pricepath towards the Black Scholes price calculated with $S_0 = 36, K = 40, r = 0.06$ and $\sigma = 0.2$

Visualized in Figure 3, it can be seen that the European CRR call price converges to the Black–Scholes price as the number of time steps increases. In 3b the deviation from the Black-Scholes price is under 0.1% after the number of time steps surpasses 400.

A similar convergence property holds for American options. As the number of discretization points increases, the Bermudan approximation becomes increasingly accurate, since the exercise dates becomes denser in time. Consequently, the discrete optimal stopping problem converges to the continuous-time optimal stopping problem underlying the American option, implying that the CRR approximation converges to the continuous-time American option price.

Even though the CRR method is both simple, computational fast and converges to the Black-Scholes formula, it is not very efficient if there are multiple stochastic processes. If there is a model where the volatility or interest rate is also stochastic, each node of the tree would be a two dimensional problem. This would increase both the complexity and the computing time of the model. Since the model has difficulties with multiple stochastic processes makes it necessary to look at a model that is able to handle more complex situations.

3.2 LSM Monte-Carlo algorithm

The Least Square Monte Carlo Method (LSM) from the Longstaff and Schwartz (2001) paper is then introduced. When fixed parameters are given the CRR method and the LSM method will return the same results. In the CRR model, continuation values are obtained directly from the risk-neutral expectation under the Q-measure. In contrast the LSM algorithm approximates this expectation through simulation.

The LSM method begins by generating a large number of stock price paths under the risk-neutral measure using Monte Carlo simulation. Starting from the final exercise date and moving recursively backwards in time, the algorithm determines whether exercising or continuing is optimal for each predetermined exercise opportunity.

For American-style options, the holder faces an optimal stopping problem at each exercise date, where the payoff from immediate exercise must be compared with the expected discounted payoff from continuing the option. The continuation value is therefore central in determining the optimal exercise strategy.

At maturity, the option value is known since the holder only chooses to exercise if the option is in the money. At earlier exercise dates, the continuation value is unknown and must be estimated by the method reviewed in Section 2.5. In Longstaff and Schwartz (2001), the continuation value is estimated using a least squares regression of the realized discounted future cash flows on functions of the underlying state variables, such as the stock price. The fitted regression function gives an approximation of the conditional expectation of the continuation value given by:

$$C_t(S_t) = \sum_{k=0}^{M} \alpha_k \cdot \mathcal{L}_k(S_t). \tag{3.1}$$

Here $L_k(S_t)$ are the Laguerre polynomial basis functions evaluated at the simulated stock price. The weighted Laguerre polynomial are defined as:

$$L_0(X) = \exp\left(-\frac{X}{2}\right), \quad L_1(X) = \exp\left(-\frac{X}{2}\right)\cdot(1-X), \quad L_2(X) = \exp\left(-\frac{X}{2}\right)\cdot\left(1-2X+\frac{X^2}{2}\right)$$

$$\text{and } L_n(X) = \exp\left(-\frac{X}{2}\right)\frac{e^X}{n!}\frac{d^n}{dX^n}(X^N e^{-X})$$

In equation 3.1, α_k are the regression coefficients estimated through ordinary least squares.

For each time step, the discounted future payoffs from continuation are regressed on a set of the Laguerre basis functions. Alternative choices of basis functions such as Hermite or Monomials are also available and discussed in Section 3.2.1. The regression is performed only on paths where the option is in the money, as the exercise decision is only relevant for these paths. Improving the computational efficiency and making the algorithm more accurate when estimating the continuation value.

After estimating the continuation value, the immediate exercise payoff is compared to the estimated continuation value for each simulated path. If the exercise payoff exceeds the continuation value, the option is exercised; otherwise, the option will remain alive. Resulting in the value process V given by:

$$V(t_n) = \begin{cases} (K - S_{t_n})^+, & h(t_n) \geq C(t_n) \\ e^{-r\,dt}V(t_{n+1}), & h(t_n) < C(t_n) \end{cases} \qquad \forall\, t_n < T.$$

In the value process $h(t_n)$ being the immediate exercise value and $C(t_n)$ is the continuation value at time t. Where the continuation value is based off regression based conditional expectation defined:

$$C(t_n) = \mathbb{E}[e^{-rdt}V(t_{n+1})|S_{t_{n+1}}].$$

This process is repeated recursively backwards through all exercise dates until the initial time is reached. Finally, the option price can be obtained as the average discounted payoff across all of the simulated paths given:

$$\Pi = \frac{1}{I}\sum_{i=1}^{I} V(0, \omega_i). \tag{3.2}$$

When choosing to implement the LSM algorithm it is to get the flexibility that is not available in the CRR model. Since the LSM relies on simulation, it can handle high-dimensional problems and more complex stochastic processes where tree-based methods become computationally infeasible. This makes the LSM method particularly useful for valuing American-style derivatives with multiple risk factors or complex payoff structures.

Table 1 shows the prices of three different models the CRR, the LSM and the European Black-Scholes. It also displays the standard error of the simulated prices for the LSM model. For comparing the the difference of the simulated prices, the table also displays the percent deviation from the CRR and the early exercise feature. The eary exercise feature is calculated as the difference from the European price to the simulated LSM price.

Table 1: Comparison of Option Pricing Methods

S	σ	T	CRR American Price	Simulated American Price	Standard Error	% Deviation	Black–Scholes European Price	Early Exercise Value
36	0.200	1	4.487	4.472	(0.009)	-0.33%	3.844	0.628
36	0.200	2	4.848	4.848	(0.011)	0.00%	3.763	1.085
36	0.400	1	7.109	7.068	(0.019)	-0.57%	6.711	0.357
36	0.400	2	8.514	8.517	(0.023)	0.04%	7.700	0.817
38	0.200	1	3.257	3.252	(0.009)	-0.15%	2.852	0.400
38	0.200	2	3.751	3.730	(0.011)	-0.56%	2.991	0.740
38	0.400	1	6.154	6.133	(0.019)	-0.34%	5.834	0.299
38	0.400	2	7.675	7.670	(0.022)	-0.07%	6.979	0.691
40	0.200	1	2.319	2.319	(0.009)	0.00%	2.066	0.252
40	0.200	2	2.890	2.900	(0.011)	0.35%	2.356	0.544
40	0.400	1	5.318	5.321	(0.018)	0.06%	5.060	0.261
40	0.400	2	6.923	6.887	(0.022)	-0.52%	6.326	0.561
42	0.200	1	1.621	1.622	(0.008)	0.06%	1.465	0.157
42	0.200	2	2.217	2.217	(0.010)	0.00%	1.841	0.375
42	0.400	1	4.588	4.561	(0.017)	-0.59%	4.379	0.182
42	0.400	2	6.250	6.254	(0.021)	0.06%	5.736	0.518
44	0.200	1	1.113	1.107	(0.006)	-0.54%	1.017	0.090
44	0.200	2	1.694	1.687	(0.009)	-0.41%	1.429	0.257
44	0.400	1	3.954	3.949	(0.016)	-0.13%	3.783	0.166
44	0.400	2	5.647	5.631	(0.021)	-0.28%	5.202	0.429

The CRR price(Benchmark), the LSM simulated price together with the standard error and the %-deviation from the benchmark alongside the Black-Scholes price and early exercise value. The LSM prices are simulated with $K = 40, r = 0.06, 50$ exercise times, $I = 100000$ and using the Laguerre polynomial. Notice the CRR has $I = 10000$.

The results in Table 1 demonstrate that the simulated Least Squares Monte Carlo (LSM) prices closely align with the benchmark prices obtained from the CRR binomial model. This is seen from the percentage deviation being under 1% for all combinations of S, σ and T. This suggests that the LSM algorithm provides a reliable approximation for pricing an American put options.

The results show a pattern. The option prices generally increase with time to maturity, reflecting the greater flexibility and opportunity for the option to move into the money over a longer horizon. For example, when $S = 36$ and $\sigma = 0.2$, the simulated American option price increases from 4.472 to 4.848 as maturity increases from $T = 1$ to $T = 2$. Another parameter choice increases the option prices is a higher volatility. This is the case since greater uncertainty raises the likelihood of favorable payoff outcomes for the holder of a put option. This effect is consistently observed across all initial stock prices and maturities.

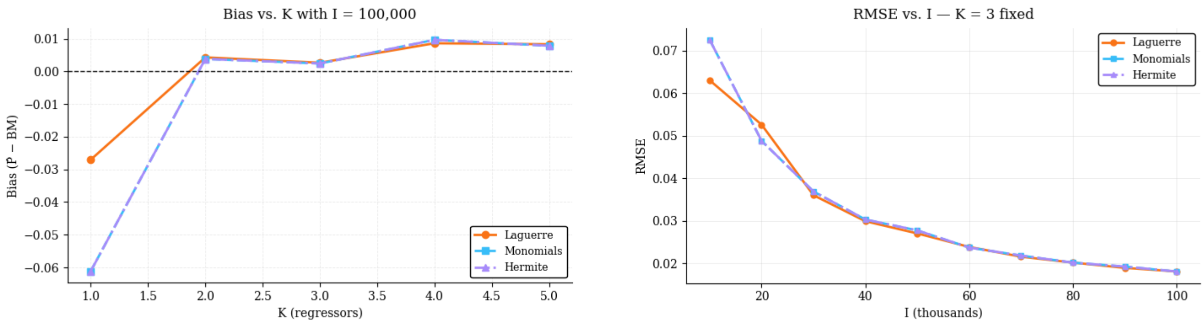
As expected for put options, there is also an inverse relationship between the underlying stock price and the option value. Holding volatility and maturity fixed, higher stock prices reduce the probability that the option finishes in the money, leading to lower option prices. For instance, with $\sigma = 0.2$ and T=1, the simulated American option price declines from 4.472 at S=36 to 1.107 at S=44.

The early exercise value, defined as the difference between the American and European option price, is positive in all cases, confirming the additional value is dependent on the early exercise feature of American options. Furthermore, this value generally increases with maturity, indicating that longer exercise windows enhance the flexibility afforded to the holder. However, the impact of volatility on early exercise value appears less systematic and depends on the degree to which the option is in or out of the money.

3.2.1 Parameter considerations

Based on some of the critique points discussed in Stentoft (2004), different approaches are taken to look into the critic of the LSM algorithm.

First we look into the choice of polynomial used for regression in the original LSM method. Originally the Laguerre polynomials are used when doing the regression, where Stentoft suggests both the Legendre/Monomials, Hermite and Chebyshev polynomials. The focus in this report will lay on the Monomials which are used when computing the Greeks and the Hermite polynomials. To see if a different choice of regression polynomial would estimate the price of an American option better, both the bias and root mean square error (RMSE) are plotted for all three choices of regressor.



(a) Bias

(b) RMSE

Figure 4: Comparing polynomial regressions

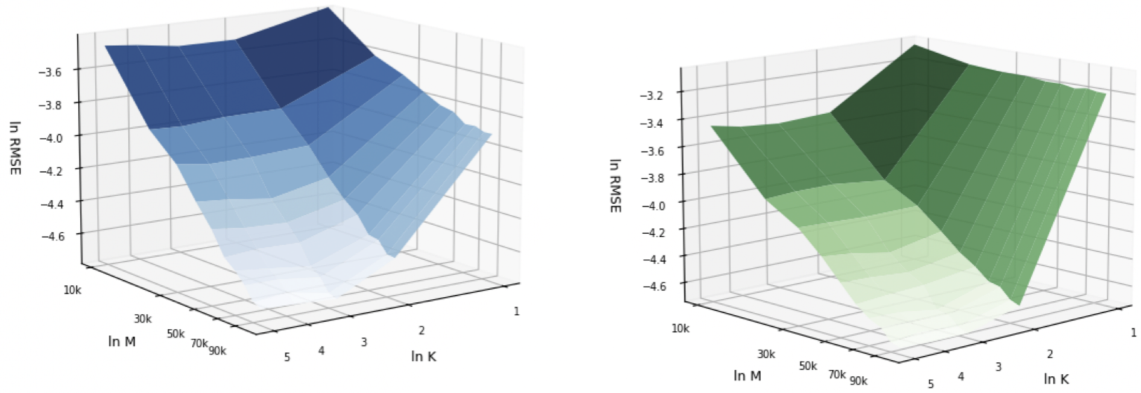
The Bias and RMSE for the three different polynomial types. All simulated with $S_0 = 40, K = 40$, $r = 0.06$ and $\sigma = 0.06$.

In Figure 4a the bias of the estimated price is plotted against the number of regressors. For $K = 1$ all three polynomials have negative bias meaning underestimating the price. With only one regressor the continuation value is poorly estimated, causing early exercise decisions that are not optimal and lowering the price. As K increases the bias converges to 0. From $K = 3$ all polynomial bases are very close to zero for all choices of K suggesting 3 regressors is sufficient.

In Figure 4b the $RMSE = \sqrt{Bias^2 + Variance}$, is plotted against the number of paths simulated, here using $K = 3$ fixed since this was found sufficient. This figure shows that the total error decreases as more simulation paths are added. When $K = 3$ is set, all three bases follow each other, which is to be expected since they were all well-specified at this point.

Synes du jeg skal lave en tabel med priserne og afvigelserne for et mere numerisk argument for at man bare kan bruge Laguerre?

Since the Hermite and Monomials follow each other closely, only the monomials and Laguerre are observed when visualizing the 3D RMSE surface.



(a) Laguerre polynomials

(b) Monomials

Figure 5: Comparing RMSE surfaces for Laguerre polynpmials and Monomials

The RMSE surface for at the money option is plotted for the LSM model. Both the LSM with Laguerre and Monomials are simulated with $S_0 = 40, K = 40, r = 0.06$, $\sigma = 0.4$ and $T = 1$.

From Figure 5 it is seen that the RMSE surface for both the Laguerre and Monomials behave similarly. The Laguerre polynomials create a smooth surface declining as both K and M increase. Where the monomials has a steep surface when moving from $K = 1$ to $K = 2$. At $K = 1$ monomials perform poorly because a single monomial is a weak regressor, but converging for $K \geq 2$.

Figure 5a shows a smoother and more stable surface across all K values. The monomials have a greater instability when looking at lower order regressions. From the plots it can be concluded that the LSM algorithm is very resilient to the choice and the number of basis function, since the bias for $K \geq 3$ is smaller than 0.01 for all three choices of basis. The Laguerre polynomial works as a suffecient and stable choice producing more reliable results across all choices of K, whereas monomials might be slightly more sensitive to this choice when is small.

Having established that the Laguerre polynomial basis with $K \geq 3$ regressors provides a sufficient approximation of the continuation value when using the LSM algorithm, we now wish to question the simulation accuracy. Even with the optimal basis, the LSM estimator is simulation based having a possibility of simulation errors, which decreases as the number of simulated paths I increases. Varying the number of paths across $I \in \{10^3, 10^4, 10^5, 10^6\}$ while computing the American put price, we report the estimated price along with a $\pm$ standard error band, which quantifies the simulation uncertainty.

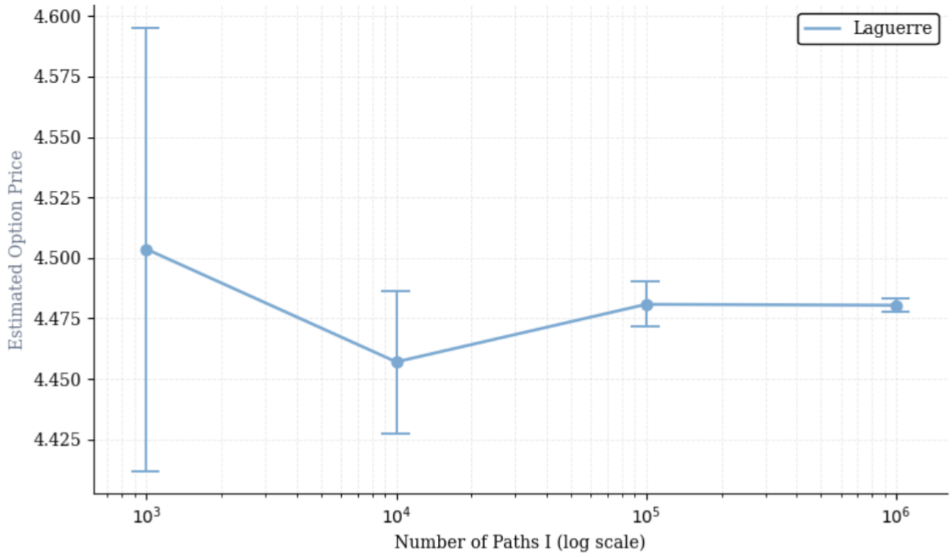


Figure 6: Estimated option price with SE against different number of paths

Stock price with ± SE band for LSM simulated prices with $S_0 = 36, K = 40, r = 0.06$ and $\sigma = 0.2$.

It is clear to see that when the number of paths (I) increases, the standard error for the estimated price gets significantly smaller, however a larger I also increasing the computation speed. When applying the extension wishing to compute the Greeks, the computation time will be relevant, and therefore the price estimate is deemed sufficiently accurate when $I = 100000$.

4 Greeks

Working with options contains more than just pricing the options using different methods. In addition to the pricing problem, it can be interesting to look at the Greeks, including Delta and Gamma. Greeks can be interpreted as the markets sensitivity to certain parameters. In this paper we wish to obtain an analytical way to estimate delta and gamma and how best to simulate accurate values of these so they can be used for hedging. Delta is the option price sensitivity with regard to changes in the underlying asset price. Gamma is the second derivative and shows deltas sensitivity to the underlying asset price. When calculated they are both rate parameters and defined as:

$$\Delta = \frac{\partial V}{\partial S}, \quad \Gamma = \frac{\partial \Delta}{\partial S} = \frac{\partial^2 V}{\partial S^2}.$$

Using the already established methods for pricing American options, different numerical methods to estimate delta and gamma are implemented and compared.

4.1 CRR

Again the CRR binomial model will work as our benchmark since it converges to the American put price as the number of time steps increases. Therefore we also need to calculate both delta and gamma using this model. This is done to be able to compare the different models implemented for calculating the Greeks. The CRR model continues to serve as a useful benchmark for calculating the Greeks due to its consistency and reliability in option pricing.

When finding the Greeks in the binomial model, they are calculated from the value function and the stock price in each node of the binomial tree. Delta is the sensitivity of the option value to the underlying stock price. In the CRR binomial model, delta is computed locally at each node using the option values in the up and down states:

$$\Delta = \frac{V_u - V_d}{S_u - S_d}$$

V_u and V_d are the option value at the the next time step after an up and down move, and S_u and S_d are the corresponding stock prices. Delta is a measure for the change in the option value based on small changes in the stock price. Gamma captures how Delta changes as the stock price shifts and measures the curvature of the option value with respect to the underlying price. This is given by:

$$\Gamma = \frac{\Delta_u - \Delta_d}{(S_u - S_d)/S_0}$$

The benchmark works as our stable estimator, since it gives the same answer no matter what seed is used, since it is not based on simulations. Even though the benchmark estimates in this paper is taken as the true values, it is necessary to look at other methods to estimate the Greeks because the CRR method can not handle complex models. When working in the binomial tree, it is hard to have more than one stochastic process. Therefore expansions of our framework, for example stochastic volatility will make the model complex and the computing time slow.

4.2 PDM and LRM

The first methods implemented for estimating the Greeks are The Pathwise Derivative Method (PDM) and The Likelihood Ratio Method(LRM) based on the revision of the models in Glasserman (2004) - chapter 7.

When implementing the PDM the advantage is, that estimates are based directly on parameters simulated in the LSM algorithm. The Pathwise Derivative Method estimates the Greeks by differentiating the payoff function for each Monte Carlo simulated path. This is done in the same style as when the CRR sensitivities are found, but here the idea is to interchange differentiation and expectation under appropriate regularity conditions. The value of the contingent claim is given by:

$$H(\theta) = E[V(Z(\theta))]$$

where $Z(\theta)$ denotes the underlying stochastic process depending on a parameter θ, and V is the payoff function. If certain smoothness and integrability conditions are satisfied, differentiation can be moved inside the expectation:

$$\frac{\partial H}{\partial \theta} = \mathbb{E}\left[\frac{\partial}{\partial \theta} V(Z(\theta))\right] \tag{4.1}$$

The equality of 4.1 ensures that it is possible to estimate the PDM greeks by differentiating the simulated payoff directly. Under the geometric Brownian motion model introduced in Section 2.3, the terminal asset price S_T is given by equation 2.1.

When focusing on American options, the payoff is realized at the optimal stopping time τ. Within the Longstaff–Schwartz framework, this stopping time is approximated numerically using regression techniques as explained in Section 2.5. The option value is then given by

$$V = \mathbb{E}[e^{-rT}\Phi(S_\tau)]$$

The path-dependent deltas are then computed following example 7.2.1:

$$\Delta = \mathbb{E}\left[e^{-r\tau}\Phi'(S_\tau)\frac{\partial S_\tau}{\partial S_0}\right] = \mathbb{E}\left[e^{-r\tau}\Phi'(S_\tau)\frac{S_\tau}{S_0}\right]$$

For an American call option, where $\Phi(S_\tau) = (S_\tau - K)^+$ the derivative of the payoff is: $\Phi'(S_\tau) = \mathbb{1}\{S_\tau > K\}$, since there only is a payoff if $S_\tau > K$. The Delta expression can then be simplified becoming :

$$\Delta = \mathbb{E}\left[e^{-r\tau}\mathbb{1}\{S_\tau > K\}\frac{S_\tau}{S_0}\right]$$

When finding the PDM Δ the pathwise deltas are computed, based off the simulations done in the LSM algorithm. Here the theoretical τ is replaced by the simulated stopping times. Lastly for getting one delta for each combination of parameters, the mean of the pathwise deltas is computed resulting in the PDM Δ. Whereas Gamma can not be estimated with this framework due to the discontinuity introduced by the simulated stopping times.

The Likelihood Ratio Method instead has the advantage that the payoffs do not have to be differentiable. The LRM provides an alternative approach to estimating sensitivities by differentiating the probability distribution of the underlying process rather than the payoff function. In this method the contingent claim is then given by;

$$H(\theta) = \mathbb{E}[V(X)] = \int V(X)f(x,\theta)dx$$

where $f(x;\theta)$ denotes the probability density of the underlying random variable X, depending on parameter θ. Using the chain rule differentiating under the integral sign gives

$$\frac{\partial H}{\partial \theta} = \int V(X)\frac{\partial f(x,\theta)}{\partial \theta}dx$$

with,

$$\frac{\partial f(x,\theta)}{\partial \theta} = f(x,\theta)\frac{\partial}{\partial \theta}\log f(x,\theta).$$

The derivative can then be written as

$$\frac{\partial H}{\partial \theta} = \mathbb{E}[V(X)\frac{\partial}{\partial \theta}\log f(X,\theta)].$$

Here $\frac{\partial}{\partial \theta}\log f(X,\theta)$ is the likelihood ratio weight. Applying this likelihood ratio to the calculation of the pathwise delta it will become:

$$\Delta = \mathbb{E}\left[e^{-rT}\Phi(S_\tau)\frac{\partial}{\partial S_0}logf(S_\tau, S_0)\right]$$

Where:

$$\frac{\partial}{\partial S_0}logf(S_\tau, S_0) = \frac{Z}{S_0\sigma\sqrt{\tau}}.$$

Inserting this into delta, the LRM delta estimator is given by:

$$\Delta = \mathbb{E}\left[e^{-r\tau}\Phi(S_\tau)\frac{Z}{S_0\sigma\sqrt{\tau}}\right]$$

By differentiating the likelihood ratio again, the LRM Gamma estimator can be obtained. In Glasserman (2004), gamma is then given:

$$\Gamma = \mathbb{E}\left[e^{-r\tau}\Phi(S_\tau)\frac{Z^2-1}{S_0^2\sigma^2\tau} - \frac{Z}{S_0^2\sigma\sqrt{\tau}}\right].$$

Both Delta and Gamma are dependent on the stopping time τ, which like in the PDM delta is estimated numerically through the LSM algorithm. Both Greeks are calculated path dependently while evaluating the likelihood ratio weights along each of the simulated paths. The PDM and LRM greeks are implemented in Python based on the estimates from the LSM implementation from Section 3.2. The Greeks are then calculated 100 times, afterwards finding the mean. With results printed in 2.

Table 2: Comparison of PDM and LRM to Benchmark

		Delta				**Gamma**			
K	Model	BM	Est.	St Dev.	% Dev.	BM	Est.	St Dev.	% Dev.
36	PDM	-0.1986	-0.1943	0.0011	-2.17%	0.0385	-	-	-
40	PDM	-0.4058	-0.3970	0.0017	-2.17%	0.0604	-	-	-
44	PDM	-0.6660	-0.6579	0.0026	-1.22%	0.0763	-	-	-
36	LRM	-0.1986	-0.1980	0.0054	-0.30%	0.0385	0.0395	0.0069	2.60%
40	LRM	-0.4058	-0.4036	0.0099	-0.54%	0.0604	0.0618	0.0135	2.32%
44	LRM	-0.6660	-0.6675	0.0109	0.23%	0.0763	0.0726	0.0259	-4.85%

PDM and LRM estimates for both Delta and Gamma and benchmark values (CRR). Also including standard deviation and the percentage deviation from the benchmark value. The LSM algorithm is executed with $s_0 = 40, K = 40, r = 0.06, \sigma = 0,2, T = 1,\ N = 50$ and $I = 100000$. All estimates are an average of 100 simulations.

The estimates of both the PDM and LRM method are relatively close to the calculated benchmark values. It is worth noticing that the standard deviation calculated for the estimates in the

LRM method are significantly bigger than for the PDM estimates. It is important to notice that the PDM method does not allow for the estimation of gamma even though the Delta estimations are both accurate and have a low standard deviation. In contrast, the LRM estimates can produce gamma but suffer from high standard deviation and a large percentage deviations for the gamma estimates. Therefore, alternative methods for computing the Greeks are considered.

4.3 LSM-ISD

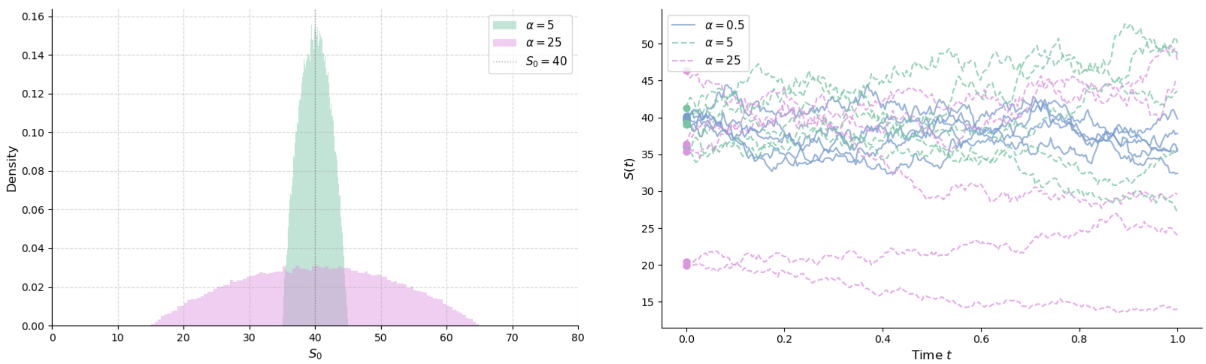
After testing both the PDM and LRM approaches, the objective is to find a single approach capable of accurately estimating both delta and gamma. In addition, the model should be flexible enough to accommodate more complex settings, such as stochastic interest rates or volatility. The Least Squares Monte Carlo Initial State Dispersion (LSM-ISD) method proposed by Letourneau and Stentoft (2023) is considered to assess whether it can provide sufficiently accurate estimates of the Greeks.

The framework in the LSM-ISD model is the same as in the LSM model reviewed in Section 3.2. In particular, the option price is obtained from the conditional expectation of discounted future payoffs under the optimal stopping rule. One of the key differences between the LSM and LSM-ISD lies in how the regression for the simulations is built and how the resulting data are used to estimate sensitivities.

In the standard LSM approach, all simulated paths are generated from a single initial value of each variables. In contrast, the LSM-ISD method introduces initial state dispersion (ISD). Here the simulation is performed from a cross-section of initial values around the point of interest. Specifically, the initial states are generated as

$$S_n = s_0 + \alpha \mathcal{K}_{ISD}(U_n) \tag{4.2}$$

Here $\mathcal{K}_{ISD}$ is the kernel density, U_n is uniformly distributed and α is the dispersion parameter.



(a) S_0 distributions for $\alpha = 5$ and $\alpha = 25$

(b) Stock paths for different dispersed S_0

Figure 7: Initial state dispersion visualizations

Here the kernel is defined as in Letourneau and Stentoft (2023) when finding the distribution and the simulations are based on the parameters $S_0 = 40, K = 40, r = 0.06, \sigma = 0.2, T = 1, N = 50$ and $I = 100000$.

In 7a the distribution of the initial stock prices at time $T = 0$ are displayed for dispersion parameters 5 and 25. It shows clearly that when α is large the initial stock price varies, but still having a bell shape centered around $S_0 = 40$. The lower the dispersion parameter is, the tighter the span of the initial stock price ends up being, pushing the bell shape both taller and slimmer. This is also why the distribution for $\alpha = 0.5$ is left out of the plot. In 7b the price paths are plotted against the time T. This also visualizes how a bigger dispersion parameter gives broader initial stock prices and can result in more extreme stock values.

Due to the assumption of the underlying process being a Markov process, simulating from different initial values does not affect the estimation of the early exercise strategy. However, it makes it possible to express the option price as a function of the initial state variable, which is important when estimating both delta and gamma. As in the standard LSM method, the continuation value at each time step is approximated using a finite set of basis functions. That is, the conditional expectation is written as a linear combination:

$$F(X(t_j)) = \sum_{m=0}^{\infty} \Phi_m(X(t_j))a_m(t_j)$$

$\Phi_m(\cdot)$ being the basis functions and $a_m(t_j)$ coefficients estimated by the regression. It is further assumed that the conditional expectation is approximated well by the first $M + 1$ terms. We can then obtain an estimate of the function as:

$$F(X(t_j)) \approx \sum_{m=0}^{M} \Phi_m(X(t_j))a_m(t_j).$$

When approximating the conditional expectation with the first $M + 1$ terms, it must hold that $m \in [0, M]$ and $j \in [1, J]$. When in the case of the LSM algorithm the price function is the average of the simulated stock values is given by equation 3.2. When looking at the LSM-ISD method, this is not the case. After finding the initial state of the stock price S_0 it is dispersed by equation 4.2. When following Letourneau and Stentoft (2023) the kernel is defined as:

$$\mathcal{K}_{ISD}(U) = 2 \cdot \sin\left(\frac{\arcsin(2 \cdot U - 1)}{2}\right).$$

In contrast to the normal LSM we now want to use another regression at time $t = 0$ and find the conditional expectation of the discounted payoff given the initial stock price; $\mathbb{E}[V(0)|S_0]$. In the conditional expectation $S_0 = \{S_0(\omega_1), ...S_0(\omega_I)\}$ is initially dispersed around s_0. Since the conditional expectations can be approximated by the first $M + 1$ terms, the price can be approximated by:

$$\hat{P}^N_{M,M_0}(S(0) = s) \approx \Pi_{M_0} = \sum_{m=0}^{M_0} \rho_m(S(0) = s))\hat{b}_m(0)$$

Here $\rho_m(S(0) = s))$ are the basis functions and $\hat{b}_m(0)$ are the coefficients estimated in the regression at time $t = 0$. In the same way both delta and gamma can be defined, since the stock price is the only state variable:

$$\hat{\Delta}^{N}_{M,M_0}(S(0)=s) \approx \Delta_{M_0} = \sum_{m=0}^{M_0} \rho'_m(S(0)=s))\hat{b}_m(0),$$

$$\hat{\Gamma}^{N}_{M,M_0}(S(0)=s) \approx \Gamma_{M_0} = \sum_{m=0}^{M_0} \rho''_m(S(0)=s))\hat{b}_m(0).$$

The regression at time t_0 can be interpreted as a local polynomial approximation of the price function around the point s_0. In particular, this corresponds to approximating the $P(x)$ using a Taylor expansion for x lying close to our s_0 as:

$$P(x) \approx P(s_0) + P'(s_0)(x-s_0) + \frac{1}{2}P''(s_0)(x-s_0)^2 + \cdots + \frac{1}{M!}P^{(M)}(s_0)(x-s_0)^M$$

Where the estimates of the derivatives is given by the solution of the OLS regression given:

$$\min_{\beta_i} \sum_{n=1}^{N} \left\{ Z_n - \sum_{i=0}^{M_0} \beta_i (S_0(\omega_i) - x_0)^i \right\}^2 . \tag{4.3}$$

Here the assumptions ensures convergence of the price, delta and gamma estimators when $\hat{P}^{(i)}(x_0) = i!\hat{\beta}_i$. Where $\beta_i, i = 0, ..., M$ are the solutions to regression problem. When the underlying stock price is the only state variable, delta and gamma are defined as:

$$\Delta = \beta_1, \quad \Gamma = 2 \cdot \beta_2.$$

The conceptual method is based on the explanation provided above. For its numerical implementation, two distinct approaches are examined.

4.3.1 LSM-ISD Naive

First introducing the naive LSM-ISD method, where the implementation can be separated into three steps. First, using the stock price dispersion defined by equation 4.2, 100000 price paths are simulated. The price is assumed to be described by a Geometric Brownian Motion, simulated from the initial dispersed stock price.

Next the LSM algorithm is applied. Here the optimal exercise boundary is used to calculate the pathwise discounted payoffs at time $t = 0$. The discounted payoffs, $V(0)$, are obtained from the cross-sectional regression in the LSM algorithm, only with in the money paths. It is important to notice as apposed to the LSM from section 3.2, the choices of I are the same for the LSM and LSM-ISD model. The same goes for both the type of basis function and the number og basis, J and M_0.

Last, the pathwise payoffs $V(0)$ are regressed on the distance of s_0 and the initial dispersed stock price, as given in equation 4.3. When completing the regression, Delta is given by the first coefficient from the regression and Gamma is 2 times the second coefficient.

When implementing the algorithm it is necessary to consider the parameter choices. One of the choices being $M_0 = 9$. From Corollary 2(Letourneau and Stentoft (2023)), it is given that it is needed to use a polynomial of at least order $M_0 > i$, in this case $i = 2$. It is shown that increasing the number of polynomials decreases the bias but increases the variance at the same time. It is wished to find the compromise between these two factors when choosing the polynomial degree. It is generally preferred to use $M_0 - i$ odd, because increasing $M_0 - i$ from even to odd will reduce the bias without effecting the variance. The case of increasing $M_0 - i$ from odd to even will also reduce the bias but increase the variance.

It is important to notice that the LSM-ISD algorithm also estimates the price, but the focus is on the Greeks, therefore the estimated stock prices will be omitted from the table, but are still calculated when running the code.

Table 3: LSM-ISD(naive) Delta and Gamma estimates

		Delta				**Gamma**			
α	K	BM	LSM-ISD	St Dev.	% dev.	BM	LSM-ISD	St Dev.	% dev.
0.5	36	-0.1986	-0.1875	0.1249	-5.59%	0.0385	0.0231	2.4814	-39.74%
0.5	40	-0.4058	-0.4187	0.2080	3.18%	0.0604	-0.1804	3.7057	-398.68%
0.5	44	-0.6660	-0.6308	0.3118	-5.29%	0.0763	-0.0422	3.9560	-155.31%
5	36	-0.1986	-0.1981	0.0116	-0.25%	0.0385	0.0382	0.0219	-0.78%
5	40	-0.4058	-0.4048	0.0190	-0.25%	0.0604	0.0544	0.0321	-9.93%
5	44	-0.6660	-0.6649	0.0219	-0.17%	0.0763	0.0792	0.0349	2.23%
25	36	-0.1986	-0.1843	0.0026	-7.80%	0.0385	0.0410	0.0008	6.49%
25	40	-0.4058	-0.4115	0.0028	1.41%	0.0604	0.0695	0.0008	15.07%
25	44	-0.6660	-0.6814	0.0038	2.31%	0.0763	0.0772	0.0010	1.18%

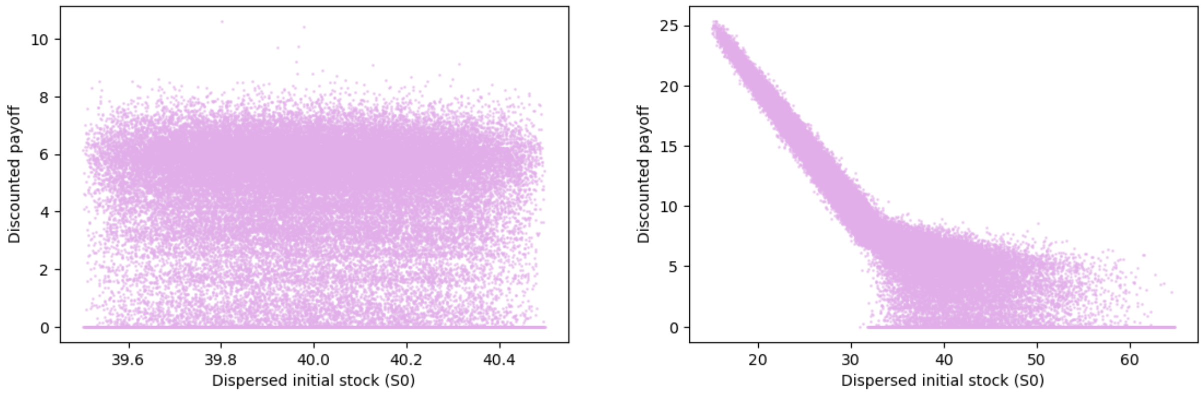
LSM-ISD(naive) estimates for both Delta and Gamma compared to benchmark values (CRR). Also

including standard deviation and the percentage deviation from the benchmark value. The LSM-ISD algorithm is executed with $s_0 = 40, r = 0.06, \sigma = 0,2, T = 1$, $N = 50, I = 100000, J = 9$ and $M_0 = 9$. All estimates are computed from an average of 100 simulations.

Looking at the results in Table 3, it is seen that there is a correlation between the dispersion parameter α and the standard deviation of the estimate. When there the dispersion parameter is small the standard deviation is significantly higher than for larger values of dispersion. This results in a higher variance. It can also be seen that for a high dispersion parameter $\alpha = 25$ the standard deviation and therefore the variance is low. When choosing α, there is a tradeoff between bias and variance. For a small dispersion parameter, the estimates have a low bias but with a higher variance. The opposite is the case for larger dispersion. It is noticeable that the % deviation for the Gamma estimates when $\alpha = 0.5$ are extremely large. This a known limitation to the naive model, and the extreme values can be due to the seed, the model is run on, and the estimates differ when running with other seeds.

Focusing on Table 3, $\alpha = 0.5$ gives high standard deviations when focusing on the gamma estimate and for both Delta and Gamma the estimates vary in their deviation to the Benchmark. The estimates with $\alpha = 25$ also deviate from the benchmark especially for the gamma estimates. Here $\alpha = 5$ seems to have estimates for both Delta and Gamma that are relatively close to the bench mark and also have a low variance.

In Figure 8 the data points used in the naive method regression are plotted for both $\alpha = 0.5$ and $\alpha = 25$.



(a) $\alpha = 0.5$ (b) $\alpha = 25$

Figure 8: Data points for LSM-ISD naive regression

The data available for regression in the LSM-ISD naive model, implemented with $S_0 = 40, K = 40, r = 0.06$ and $\sigma = 0.2$.

Seeing the distribution of the data points, it can be seen that there is no clear structure, especially for $\alpha = 0.5$. When fitting the polynomials to this data, it will create a higher bias and variance in particular for the higher order derivatives like the Gamma. This is also what can be observed in Table 3.

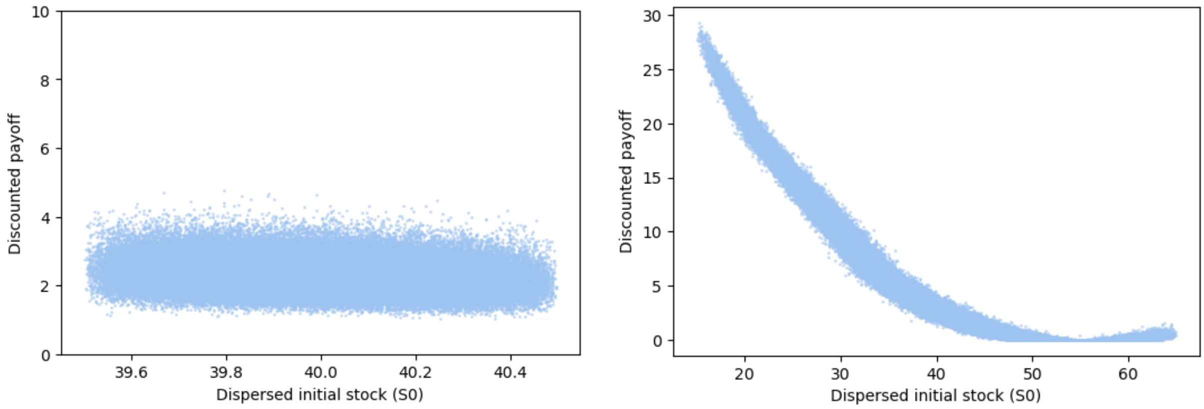
4.3.2 LSM-ISD 2 stage

After looking into the naive implementation of the LSM-ISD, it was seen that the model is sensitive to the choice of α, especially for small values, impacting the Gamma estimates. Another weakness the model shows is the high bias for both $\alpha = 0.5$ and $\alpha = 25$. Therefore the LSM-ISD Two-Stage model is introduced. Here the desired outcome is to make the model variance-reducing and also wanting to estimate the optimal dispersion parameter to get accurate estimates for the sensitivities.

After seeing the distribution of the data points that are used for the regression in the naive model, the first step is to do a variance reduction of these points. In the naive implementation we regress from $t = 0$, mixing near information and information known far out in the future. Instead of using realized discounted payoffs like in the naive implementation, we replace them with the value function at t_1 to reduce variance, given:

$$V(t_1) = \begin{cases} h(t_1), & \text{if } h(t_1) \geq \hat{C}_J(t_1) \\ \hat{C}_J(t_1), & \text{if } \hat{C}_J(t_1) < h(t_1) \end{cases} \tag{4.4}$$

The data points after the variance reduction method can be visualized in Figure 9. er det defineret ellers DEFINER h og C



(a) $\alpha = 0.5$

(b) $\alpha = 25$

Figure 9: Data points for LSM-ISD 2 stage regression

The data available for regression in the LSM-ISD 2 Stage model, implemented with $S_0 = 40, K = 40, r = 0.06$ and $\sigma = 0.2$.

Here it can be seen that the data points do not vary in the same degree/scope as before the variance reducing method. There is now an expectation that it would be possible to calculate the Greeks so they are close to the benchmark values meanwhile having a smaller standard deviation and then also variance.

The other improvement is based on finding the optimal dispersion parameter. Here the focus is to find the trade off between bias and variance, wanting the most accurate estimations. A fixed α^* is chosen, forming the interval $[-\alpha^*, \alpha^*]$. Paths that were initialized with values outside of this interval are simulated again with the α^*. In the second stage all paths will then lie within the optimal bandwidth. It is also possible to truncate the paths, discarding the outlying paths. This approach would instead lead to increased variance in the estimates and is therefore omitted.

The LSM-ISD 2-stage method can then be explained by the following steps: Like in the naive method, the first thing to do is, simulating 100000 price paths using the stock price dispersion defined by equation 4.2. The price is assumed to be described by a Geometric Brownian Motion, simulated from the initial dispersed stock price.

The LSM algorithm is then used to estimate the optimal stopping time. Using this, the pathwise discounted payoffs for time $t = 1$ are calculated. And then storing the optimal stopping times, so they can be used for the final regression.

Then the value function for time $t = 0$ is determined using the value function defined in equation

4.4. The exercise value can then be discounted to $t = 0$, "smoothing" the payoffs.

From the choice of α^*, the paths outside of the interval are re-simulated, now using a dispersion from α^*. The new simulated paths now replace the outlying paths.

Last, the pathwise payoffs for time $t = 0$ are regressed on the distance of s_0 and the dispersed stock price, as given in equation 4.3. When completing the regression, Delta is given by the first coefficient from the regression and Gamma is 2 times the second coefficient.

The estimations are displayed in Table 4 below:

Table 4: LSM-ISD Delta and Gamma estimates

		Delta				**Gamma**			
α	K	BM	LSM-ISD	St Dev.	% dev.	BM	LSM-ISD	St Dev.	% dev.
0.5	36	-0.1986	-0.1914	0.0138	-3.63%	0.0385	0.0254	0.0130	-34.03%
0.5	40	-0.4058	-0.4107	0.0055	1.21%	0.0604	0.0589	0.0068	-2.48%
0.5	44	-0.6660	-0.6679	0.0064	0.29%	0.0763	0.0713	0.0098	-6.55%
5	36	-0.1986	-0.2062	0.0034	3.83%	0.0385	0.0390	0.0032	1.30%
5	40	-0.4058	-0.4119	0.0045	1.50%	0.0604	0.0597	0.0063	-1.16%
5	44	-0.6660	-0.6719	0.0064	0.89%	0.0763	0.0761	0.0084	-0.26%
25	36	-0.1986	-0.1998	0.0029	0.60%	0.0385	0.0369	0.0032	-4.16%
25	40	-0.4058	-0.3971	0.0060	-2.14%	0.0604	0.0566	0.0062	-6.29%
25	44	-0.6660	-0.6617	0.0061	-0.65%	0.0763	0.1010	0.0111	32.37%

LSM-ISD(2-stage) estimates for both Delta and Gamma compared to benchmark values (CRR). Also including standard deviation and the percentage deviation from the benchmark value. The LSM-ISD algorithm is executed with $S_0 = 40, r = 0.06, \sigma = 0,2, T = 1, N = 50, I = 100000, J = 9$ and $M_0 = 9$. All estimates are computed from an average of 100 simulations.

Comparing the estimates in Table 4 and Table 3, it is seen that there overall is a significantly smaller standard deviation in Table 4, also making the variance lower for the 2 stage estimations. In addition to this all initial values of α come relatively close to the benchmark values. After applying the 2-stage method, there is no longer a clear pattern between initial dispersion and variance. This is a sign that the variance reducing method has worked. But it is interesting to notice that the Delta estimates for $\alpha = 5$ are now worse in the 2 stage compared to the naive model, introducing doubt wheter or not this model will perform better, for all choices of α. In the implementation $\alpha^* = 5$ was chosen manually and not computed. In the Letourneau and Stentoft (2023) paper, they introduce an alternative way to choose α^* in Appendix A.3, to further improve the model. Since the estimates from the LSM-ISD 2-Stage model with a fixed α^* seems to yield relatively accurate estimates with low standard deviation and bias, the further improvement is not implemented.

5 Delta Hedging

When there is only one source of risk introduced when pricing an option, in our case the stochastic stock price, dynamic delta hedging can be used for minimizing the risks that are dependent on the price of the underlying asset. This is possible since the greeks examined, Delta and Gamma, are the price sensitivities. When dynamic delta hedging the point is to hedge against the price changes of derivative. To do dynamic delta hedging, the goal is to delta-neutralize the price sensitivity. This can be done by fulfilling the equation:

$$dV_t - \Delta_t^V dS_t = 0$$

Reintroducing the replicating portfolio, a potential investor would like to replicate the payoff of the value function:

$$V_t = \Delta_t^V S_t + \gamma_t B_t \tag{5.1}$$

where $\gamma_t B_t$ is the position in the risk-free asset. When looking at put options, it generally implies being short of the underlying meaning $\Delta_t^V < 0$ and long the risk free asset. In continuous time the replicating portfolio must be adjusted at every instant the stock price moves. In practice, re-balancing can only occur at every discrete time step. The portfolio will then be re-evaluated, with the goal of fulfilling equation 5.1. The goal is to make a self-financing portfolio, such that it is rebalanced without any external cashflow. When working with American options, there is also the possibility of early exercise is there.

To be able to compare the Delta estimates from the implemented methods, the hedging error is measured. The hedging error is defined as the difference between the value of the self-financing replicating portfolio and the options payoff at the time of maturity. Since American options can be exercised at different points in time across paths, each hedge error is discounted back to $t = 0$ using the risk-free rate, so that all errors are expressed in present value terms and directly comparable across each simulated path.

Using the method from Hilpisch (2015) chapter 13, it would require that Delta was calculated a maximum of N_0 times for every simulated path. When looking at $I = 100000$ paths, simulating Delta and then the dynamic delta hedging would end with a very time consuming simulation process. To optimize the computation time, interpolations are used instead.

maybe interpolation-variance plots for the choice of interpolations points being 20

5.1 CRR Delta Errors

The hedging performance of the CRR-based dynamic hedging strategy is first examined, to ensure this can work as a benchmark.The performance of the CRR hedge is visualized through two plots: the distribution of discounted hedge errors across all simulated paths and the replicating portfolio relative to the option value along one representative path.

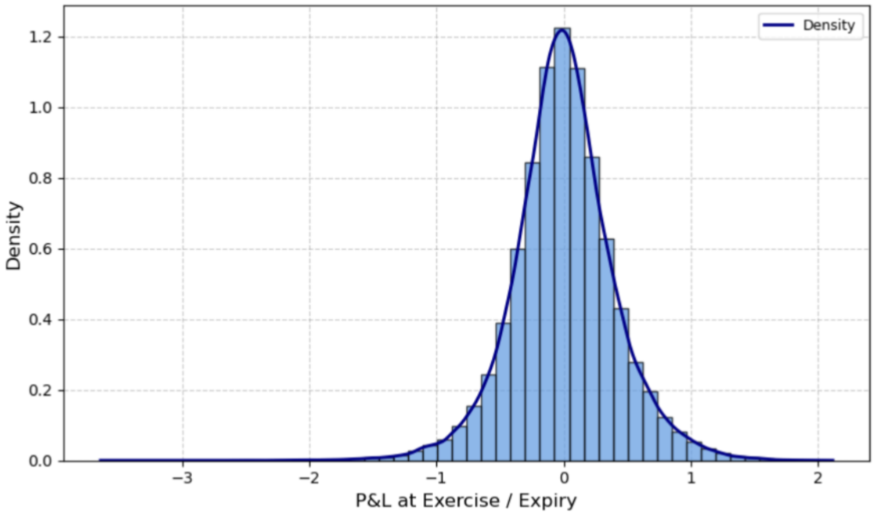


Figure 10: CRR P&L distribution

Hedging error for the CRR model visualized for $S_0 = 40, K = 40, r = 0.06$ and $\sigma = 0.2$.

A good hedge should replicate the option payoff as closely as possible, the distribution is then expected to be centered around zero. This is also shown to be the case in Figure 10 where the 99% profit and loss interval goes from $[-1.23, 1.20]$. The variance being 0.1620 is an indication for the hedging precision, where a narrower distribution implies lower replication error, which is what is observed, suggesting that the CRR hedge replicates the option payoff with relatively small deviations across different paths.

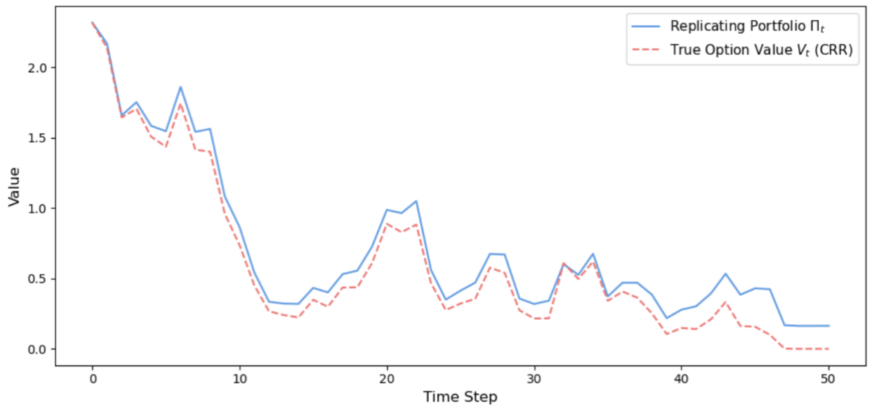


Figure 11: Comparison of the true value and replicating portfolio

Single CRR path visualized against the true option value. The paths are simulated with $S_0 = 40, K = 40, r = 0.06$ and $\sigma = 0.2$.

To illustrate the pathwise behavior of the replication strategy, Figure 11 compares the value of the replicating portfolio with the corresponding option value along a representative stock path. For the CRR benchmark, the replicating portfolio lies slightly above but following the option value, indicating that the hedge captures the dynamics of the underlying option reasonably well but slightly over hedging by keeping larger amounts in the risk free asset.

Together the plots justify the choice of using the CRR model as benchmark values.

5.2 American Delta Errors

After comparing the Greek estimations of the four different methods implemented, both the PDM and LRM model are not taken into account when comparing the Dynamic delta estimation. Therefore the LSM-ISD and LSM-ISD 2Stage methods delta estimations are compared with each other and relative to the benchmark. To compare the models best, the hedging error(i.e the P&L) is found for each price path simulated, afterwards taking the mean, the variance and then finding the 99%-interval wherein the profit and loss lay. Taking the 99%-interval ensures a better understanding of the distribution, where an interval based on the min and max will just give the potential outliers in the particular model. The results are shown in Table 5.

Table 5: Hedging with different models

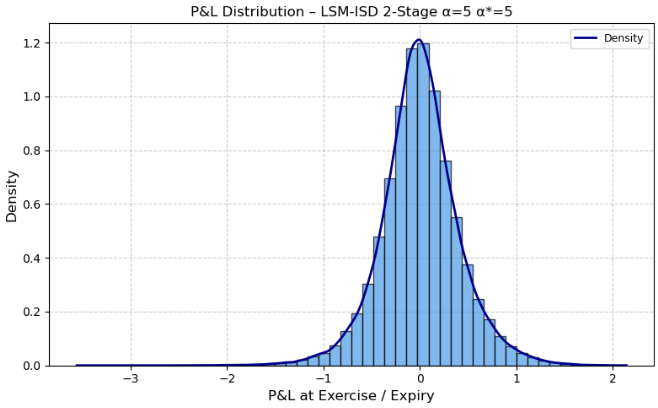
Model	Mean	Variance	P&L Interval
CRR(Benchmark)	0.0055	0.1599	[-1.24,1.20]
LSM-ISD naive α=0.5	-0.0070	0.1899	[-1.33,1.30]
LSM-ISD naive α=5	0.0181	0.1603	[-1.22,1.23]
LSM-ISD naive α=25	-0.0294	0.2436	[-1.40,1.54]
LSM-ISD 2-Stage α=0.5	0.0037	0.1603	[-1.23,1.21]
LSM-ISD 2-Stage α=5	0.0002	0.1610	[-1.24,1.21]
LSM-ISD 2-Stage α=25	-0.0027	0.1676	[-1.23,1.27]

Mean, variance and 99%P&L interval. The models are implemented with S_0 = 40, K=40, r = 0.06, σ = 0.2, T = 1, N = 50, I = 100000, M_0 = 9, α^* = 5.

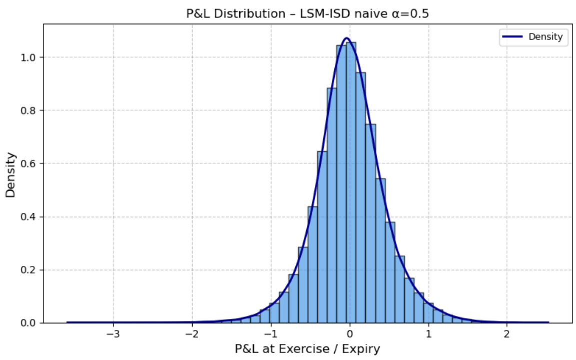
From the intervals it can be seen that the LSM-ISD naive still is dependent on the choice of α, where the LSM-ISD 2 Stage has more consistant estimates non dependent on the choice of α. Although it can be seen that the mean hedging errors are all near zero, suggesting little systematic profit or loss at expiry.

The LSM-ISD 2 stage with dispersion 0.5 and 5 perform almost identical to the benchmark interval with a slightly larger mean and variance meanwhile having a slimmer interval. Therefore it is hard to tell if the LSM-ISD 2 stage model is better when prioritizing accuracy in delta hedging. It is on the other hand noticeable that the intervals for the naive model vary a bit more. With the $\alpha = 5$ all three estimations are close to the benchmark and the 2 stage results. For both $\alpha = 0.5$ and $\alpha = 25$ the P&L intervals are wider while also having a larger variance, so even though the mean is approximately 0 hedging with this model and dispersion parameter will create more uncertainty. That is also what is expected when looking at the results in Table 3.

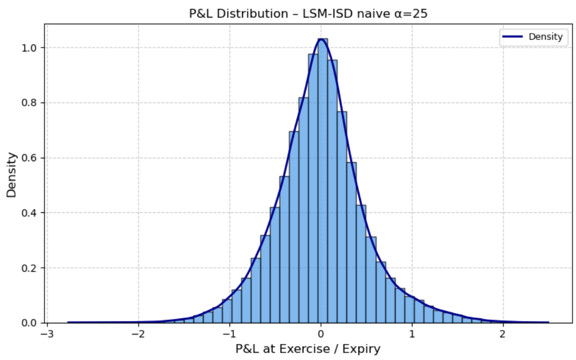
Since the estimations for the 2-stage model are very similar together with the estimation from the naive model with $\alpha = 5$, only one of these P&L distributions are shown i Figure 12. Together with the LSM-ISD naive distributions for $\alpha = 0.5$ and $\alpha = 25$ to visualize their more skewed distribution. (The rest of the distributions are plotted and can be found in Appendix 6.1.)



(a) LSM-ISD 2 stage distribution with $\alpha = 5$



(b) LSM-ISD distribution for $\alpha = 0.5$



(c) LSM-ISD distribution for $\alpha = 25$

Figure 12: LSM-ISD (naive and 2-stage) model distributions

Hedging error for the LSM-ISD model both the naive and 2-stage visualized for $S_0 = 40, K = 40, r = 0.06$ and $\sigma = 0.2$.

A common feature across all P&L distributions is their approximately symmetric, bell-shaped form centered near zero, indicating that the hedging strategies are broadly unbiased.

The two-stage model with $\alpha = 5$ produces a concentrated distribution with moderate tails and low dispersion, suggesting more stable hedging performance and a lower likelihood of extreme outcomes. This is also the case when looking at $\alpha = 0.5$ and $\alpha = 25$, see the distributions in Appendix 6.1.

For the naive model, $\alpha = 0.5$ yields a distribution centered near zero but with slightly heavier tails, indicating a little sensitivity to extreme outcomes and less stable hedging errors. Increasing the parameter to $\alpha = 25$ results in a visibly broader distribution with more pronounced tails, implying higher variability and greater hedging risk despite the mean remaining close to zero.

Overall, the results suggest that the dispersion parameter affects the shape and risk profile of the naive P&L distribution. The case $\alpha = 5$ produces tighter and more stable outcomes, consistent with the findings in Table 5, while larger values increase tail risk. The two-stage LSM-ISD model generally delivers more concentrated and risk-controlled outcomes than the naive approach since it is dependent on the choice of dispersion.

Intuitively, one may wonder how the CRR benchmark can perform as well as, or even better than, the LSM-based models even when it is not explicitly incorporating the early exercise feature of American options. The hedging performance depends on the accuracy and stability of the estimated deltas, rather than how realistic the modeling decisions are, like the early exercise feature. Since the CRR model derives hedge ratios through a no-arbitrage replication framework it results in stable delta estimates. In contrast, LSM-based methods estimate the delta through simulation and regression estimated continuation values. This can lead to approximation errors and noisier Deltas. Since early exercise is typically optimal only in specific regions of the state space, omitting this feature may have only a limited impact on overall hedging performance. Which is also observed both when looking at the distributions in Figure 10, Figure 12 and the results in Table 5.

5.3 Gamma

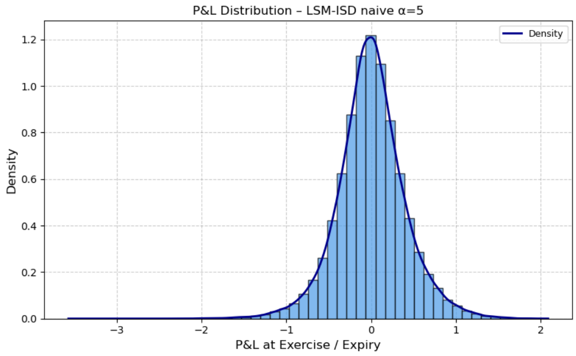
Gamma is the second order derivative, explaining deltas sensitivity to changes in the underlying price. So when delta hedging it is not enough for the replicating portfolio to be delta-neutral, since the value of Gamma effects how often re-balancing should take place. When Gamma is different from 0, Delta will change as the price of the underlying changes. Therefore it is also important to look into Gamma when doing the dynamic delta hedging. A larger Gamma will result in Delta changing at higher rates and vice versa.

6 Conclusion

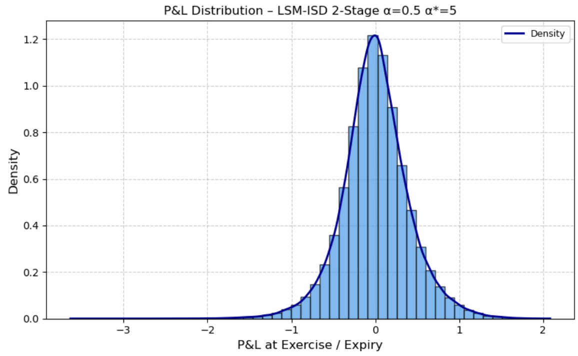
7 Appendix

LRM Delta and Gamma Deriving $\frac{\partial}{\partial S_0} log f(S_T, S_0)$:

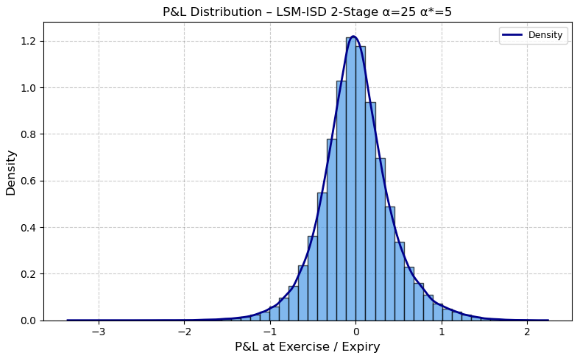
7.1 P&L distributions



(a) LSM-ISD naive distribution with α=5



(b) LSM-ISD 2 stage distribution with α=0.5



(c) LSM-ISD 2 stage distribution with α=25

Figure 13: Remaining LSM-ISD distributions

PDM OG LRM HEDGING:

Table 6: Hedging with different models

Model	Mean	Variance	P&L Interval
CRR(Benchmark)	0.0055	0.1599	[-1.24,1.20]
PDM	-0.0038	0.1591	[-1.22,1.20]
LRM	-0.0023	0.1608	[-1.24,1.20]

Mean, variance and 99%P&L interval. The models are implemented with S_0 = 40, K=40, r = 0.06, σ = 0.2, T = 1, N = 50, I = 100000, M_0 = 9, α^* = 5.

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